

THERESA, Antarctica

WMO: 893140

Lat: 84.600S

Lon: 115.817W

Elev: 1463

StdP: 84.94

Time Zone: -8.00 (W08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -46.6	(c) -43.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 23.3	(k) -29.7	(l) 20.8	(m) -30.8	(n) 5.8	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 12	(b) 4.8	(c) -5.0	(d) N/A	(e) -6.5	(f) N/A	(g) -7.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.4	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.1	(b) 19.1	(c) 17.3	(e) -49.9	(f) -0.7	(g) 2.5	(h) 2.7	(i) -51.7	(j) 1.3	(k) -53.2	(l) 2.9	(m) -54.6	(n) 4.4	(o) -56.4	(p) 6.4		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-22.6	-11.8	-17.7	-22.9	-25.8	-26.0	-27.8	-29.7	-29.1	-28.2	-23.4	-17.3	-11.4	
	DBStd	8.11	3.08	4.09	5.47	5.11	5.84	6.21	6.34	6.23	6.34	4.78	3.66	2.65	
	HDD10.0	11903	674	777	1021	1073	1117	1133	1232	1212	1145	1035	819	664	
	HDD18.3	14944	933	1010	1279	1323	1375	1383	1490	1471	1395	1293	1069	923	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	9.5	6.4	7.8	9.3	10.0	11.0	10.1	10.5	11.1	10.8	10.4	9.4	6.7	
Precipitation	PrecAvg	295	11	11	17	20	32	32	38	38	41	27	18	10	
	PrecMax	451	27	23	35	52	60	60	96	74	77	62	56	27	
	PrecMin	169	1	5	4	5	7	11	11	14	7	5	3	2	
	PrecStd	66	7	5	9	12	12	13	19	17	19	14	13	6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.2	-7.8	-9.5	-12.6	-10.1	-12.4	-14.1	-14.2	-12.2	-11.6	-7.6	-2.9	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	DB	-4.6	-9.7	-11.4	-15.5	-15.2	-15.2	-17.7	-16.2	-15.2	-13.6	-9.6	-4.9	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	DB	-6.1	-10.8	-13.2	-17.6	-17.2	-17.4	-19.5	-18.9	-17.5	-15.2	-11.0	-6.3	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	DB	-7.4	-12.1	-15.3	-19.0	-18.8	-19.4	-21.4	-20.8	-19.8	-16.8	-12.3	-7.7	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.6	5.1	5.5	5.8	5.9	6.1	6.5	6.0	4.8	4.4	4.8	
		MCDBR	6.8	4.9	5.4	6.2	6.8	7.7	6.9	7.1	7.7	5.7	5.6	7.3	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.181	0.163	0.106	N/A	N/A	N/A	N/A	N/A	0.102	0.155	0.171	0.180	
	taud		2.277	2.231	1.939	N/A	N/A	N/A	N/A	N/A	1.893	2.193	2.270	2.287	
	Ebn at Noon		995	902	643	0	0	0	0	0	623	913	1009	1028	
	Edh at Noon		79	56	24	0	0	0	0	0	24	59	79	85	
All-Sky Solar Radiation	RadAvg		8.59	4.69	0.87	0.00	0.00	0.00	0.00	0.00	0.26	3.15	7.68	10.09	
	RadStd		0.24	0.19	0.04	0.00	0.00	0.00	0.00	0.00	0.02	0.12	0.29	0.33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page